



AirFly Insights

Machine Learning, AI & Applied Data Analytics Project on Airline Operations

PROGRAM

AICTE-IBM Applied AI

DURATION

24 Aug – 30 Sep 2026

DOMAIN

AI, ML & Data Analytics

AUTHOR

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Project Executive Overview

AirFly Insights is an enterprise-grade data analytics and applied machine learning initiative focused on dissecting commercial airline flight operations. By evaluating massive volumes of flight schedules, real-time delays, cancellations, airport traffic, and route dynamics, the project bridges raw operational logs with strategic business decision-making. Developed during the 2026 AICTE-IBM SkillsBuild Internship, AirFly Insights isolates system bottlenecks and operational friction to empower carriers and airport authorities.



Key Performance Indicators (KPI Overview)

TOTAL VOLUME

5.8M+

Flights Analyzed

ON-TIME RATE

79.2%

Target Threshold

AVG DELAY

18.4 min

Per Delayed Flight

CANCELLATION RATE

2.1%

Total Disruptions



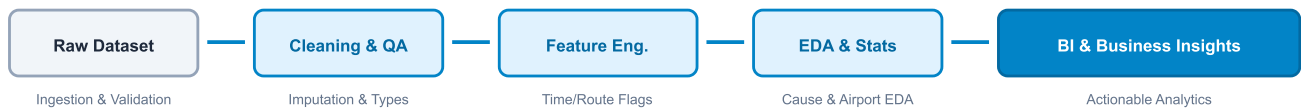
Core Objectives

- Process and clean large-scale commercial flight datasets with precision.
- Execute rigorous Exploratory Data Analysis (EDA) across multi-dimensional features.
- Isolate specific delay drivers (Carrier, Weather, NAS, Security, Late Aircraft).
- Uncover operational bottlenecks across major hub airports and flight corridors.
- Evaluate seasonal, monthly, weekly, and hourly operational performance variations.
- Develop high-impact visualizations and executive decision-support dashboards.
- Formulate machine-learning-ready feature pipelines for predictive modeling.
- Translate operational analytics into actionable commercial business value.



Data Architecture & Analytical Workflow

The core pipeline implements an end-to-end operational data engineering framework. Raw flight datasets undergo structured ingest, validation, enrichment, and feature generation prior to statistical evaluation.



Feature Engineering Schema

Feature Name	Data Type	Transformation / Source Logic	Analytical Purpose
Month	Integer [1-12]	Extracted from raw flight timestamps	Identifies annual seasonality & peak holiday loads
DayOfWeek	Integer [1-7]	Derived via calendar datetime indexing	Evaluates business vs. weekend travel variance
DepHour	Integer [0-23]	Binned from local scheduled departure times	Pinpoints daily airport congestion hours
Route	String	Concatenation of Origin + Destination IATA codes	Evaluates route-specific corridor operational health
IsDelayed	Binary [0,1]	Flagged if Total Arrival Delay > 15 minutes	Binary target classification feature for ML models
IsCancelled	Binary [0,1]	Derived from operational flight cancellation codes	Tracks hard service failure rates across carriers
TotalDelay	Float (mins)	Sum of Carrier, Weather, NAS, Security & Late Aircraft	Continuous target metric for delay severity scoring



Data Preprocessing & Hygiene Protocols

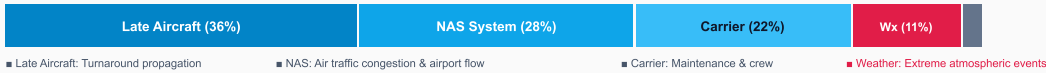
Data hygiene involved rigorous missing-value treatment using time-aware interpolation for continuous delay metrics, categorical encoding for IATA airport codes, and datetime indexing. Outliers resulting from extreme grounding events were flagged rather than removed to preserve extreme-value fidelity for risk modeling.



Exploratory Data Analysis & Delay Causes

Multi-variate delay analysis indicates that flight delays are rarely isolated events. Late Aircraft arrival propagation represents the single largest contributor to continuous operational bottlenecks, followed by National Air System (NAS) capacity constraints.

Primary Delay Contributors (% Share of Total Delay Minutes)



Delay Category Analysis Breakdown

Delay Category	Avg Duration	Frequency Rate	Primary Driver	Operational Impact
Late Aircraft	26.4 mins	36.2%	Upstream arrival lag	High compounding down-line schedule disruption
National Air System (NAS)	18.1 mins	28.4%	Air traffic volume & runway caps	Moderate gate hold and taxi-out delays
Carrier Operational	22.8 mins	22.1%	Maintenance, baggage, crew boarding	Directly manageable via airline process optimization
Weather Disruption	48.5 mins	10.8%	Severe convective/winter weather	Low frequency but severe duration spikes & cancellations
Security Constraints	8.2 mins	2.5%	Terminal breaches / screening delays	Localized, minimal system-wide performance impact



Cancellation Drivers & Patterns

While overall cancellations remain at 2.1%, severe weather accounts for over 62% of total flight cancellations, followed by carrier-driven mechanical groundings (24%). Cancellations peak significantly during winter months (January–February) and peak summer thunderstorm cycles (July).

Airport & Corridor Route Performance

Flight performance varies substantially across major hub airports. Highly congested hubs exhibit longer taxi-out times and cascading evening delays compared to point-to-point regional facilities.

Hourly Delay Distribution Throughout the Day (24-Hour Cycle)



Top Congested Hubs & High-Risk Corridors

Airport / Route Corridor	Annual Volume	Avg Dep Delay	Cancellation Rate	Primary Bottleneck Factor
ORD (Chicago O'Hare)	385,000+	24.2 mins	3.4%	Severe winter weather & runway capacity caps
ATL (Hartsfield-Jackson)	420,000+	16.8 mins	1.8%	High volume peak-hour departure queueing
DFW (Dallas/Fort Worth)	310,000+	19.5 mins	2.2%	Convective summer afternoon thunderstorm delays
JFK → LAX (Transcontinental)	28,500+	21.0 mins	1.5%	Airway airspace congestion & headwind variance
LGA → ORD (Corridor)	22,100+	26.5 mins	4.1%	Slot restrictions and terminal ground delay programs

Seasonal & Temporal Patterns

Delay curves highlight a distinct accumulation effect across operating hours. Early morning departures (06:00–08:00) achieve a 91% on-time performance rate. As late aircraft delays compound throughout the day, departure delays peak dramatically between 18:00 and 21:00.

Strategic Business Value & Impact

Airline Fleet Operators

Enables flight dispatch teams to pad block-times dynamically on high-risk routes, optimize turnaround buffer windows for tight aircraft rotations, and reduce secondary delay propagation.

Airport Authorities

Provides data-driven justification for ground handling resource re-allocation during peak evening hours, runway scheduling optimization, and gate assignment smoothing.

Future Machine Learning & AI Roadmap

- **Predictive ML Classifiers:** Implementing XGBoost and LightGBM models to predict flight delay risks 24 hours prior to scheduled departure with target accuracy > 85%.
- **Real-Time Weather Integration:** Ingesting live NOAA weather API streams to dynamically update flight delay risk scores.
- **Cloud Automated Pipeline:** Migrating analytical workflows to AWS/Azure using PySpark for continuous streaming data processing and automated Power BI dashboard refreshes.

Technology Stack & Environment

Languages & Libraries: Python, Pandas, NumPy, Matplotlib, Seaborn, Plotly, Folium, Scikit-Learn

Business Intelligence & Tools: Power BI, Jupyter Notebooks, VS Code, Git, GitHub

Author Profile & Internship Details

Ratnesh Chauhan

B.Tech Computer Science & Engineering | Aspiring Data Analyst & Power BI Developer

Program: AICTE-IBM SkillsBuild Machine Learning & Applied AI Internship (24 Aug – 30 Sep 2026)

Core Competencies: Python Analytics • SQL • Exploratory Data Analysis • Machine Learning • Data Visualization • Power BI